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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Lumina Nusantara
Project / Solution Name	SkillRadar MY
Name (Leader)	DU YULIN
Email & Phone (Leader)	CYS2609009@xmu.edu.my / +60 11-1798 1756
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app
Code Repository URL (public / access granted to judges)	https://github.com/douyinlin/sdc2026-lumina-nusantara (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Synthetic profile or questionnaire -> Streamlit interface -> local skill extractor -> five-component matching engine -> learning pathway and SDG display. The engine reads team-curated JSON catalogues and dated trend snapshots from the repository; a human reviews new source material and runs the validation/import command. There is no database, no external inference service and no GPU on the request path. The public demo runs on Streamlit Community Cloud from the public GitHub repository; Replit is documented as the development workspace.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Streamlit 1.66.0	Web interface and session state	Apache License 2.0
pandas 3.0.6	Tabular handling for roles, skills and course records	BSD 3-Clause
NumPy 2.5.4	Numeric helpers for charts and scoring	BSD 3-Clause
PyArrow 26.0.0	Columnar serialisation used by pandas	Apache License 2.0
Matplotlib 3.11.2	Occupational radar chart on the results page	Matplotlib licence (BSD-style)
ReportLab	Build-time generation of the submission PDFs	BSD 3-Clause

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None	The running application calls no external API; CV parsing and scoring are local.	-

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Streamlit Community Cloud	Hosts the public demo (auto-deploy from the GitHub main branch)	Free tier / default region
GitHub	Source repository and submission folder	Free (public repository)
Replit	Documented development workspace; screenshots of use in the repository	Free workspace

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	-	-	No model is embedded in or called by the application

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM Job Market Insights, 2025 Q1/Q2 and 2026 Q1/Q2	https://www.dosm.gov.my (release pages, listed in data/trends)	public	Government publication, cited with links	N
JobStreet by SEEK / Randstad 2025 skill signals	https://theedgemalaysia.com/node/791561	public	News article, quoted briefly with attribution	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
WEF Future of Jobs Report 2025 - Skills Outlook	https://www.weforum.org/publications/the-future-of-jobs-report-2025/3-skills-outlook	public	WEF publication, cited with link	N
Team-curated role and skill catalogue (45 roles, 54 skills)	repository: data/roles.json, data/skills_sdg.json	self-collected (team build)	Project Apache License 2.0	N
Synthetic sample CV and demo profiles	repository: data/sample_cv.txt, data/demo.json	synthetic	Project Apache License 2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
DeepSeek (chat assistant)	Drafting and reviewing code, copy and documentation	major
GitHub Copilot (code completion)	Inline code suggestions during development	moderate
OpenAI Codex (agentic coding assistant)	Refactors, regression tests and documentation edits	moderate
ChatGPT	Wording of interface copy and document drafts	minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-assisted change was read, run and tested by a team member before it was accepted: the unit and page tests must pass, the flow is exercised in a browser with synthetic data, and anything the team could not explain was rewritten or removed. AI tools produced drafts, not decisions. The trend numbers are transcribed from published releases by hand and each row keeps its source link.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☒ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
None	-	No sponsor-provided resource was used in this project.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not applicable. No sponsor-provided API, credit or dataset is used, so nothing in the prototype depends on a sponsor resource. Scoring, parsing and data loading are entirely local.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application collects no personal data and stores nothing on a server. Demo profiles and the sample CV are synthetic and shipped in the repository; the CV uploader parses a file in memory and writes only derived skill labels into the Streamlit session. Nothing is sent to a third party, because no external API is called. Users are told not to upload real CVs. Clearing the session discards the parsed profile.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The mapping from an occupation group to a skill is a team heuristic, not an official crosswalk, so trend input is coarse. Skill extraction is keyword and alias matching: it misses synonyms and cannot read scanned PDFs. Course hours and prices were not re-verified, and some skills have no verified course at all. Scoring is a ranking rule, not a hiring probability, and the two scenarios are estimates. A single quarter of DOSM data is not a trend; the interface says so.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

Leader signature:



Name: DU YULIN

Actual signing date: 24 October 2026